Parity-Separated Three-Root Norm
Thirteen Integers, Degree 156, 127 Gates

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1 Parity-separated roots: thirteen integers, degree 156

This endpoint uses thirteen unrestricted integers, has exact degree 156, and has a division-free evaluator of 127 binary operations: 75 multiplications, 27 additions and 25 subtractions. It eliminates both the strong Pell root and the triangular positivity coordinate from the fixed fifteen-variable free-scale source. Its zero existence is equivalent to that source by maps on every zero, not just a canonical subclass. The same ordinary source34 compiler/calibration therefore supplies the ZFC-provable equivalence between zero nonexistence and syntactic Con(ZFC); this is not a proof of consistency or a new end-to-end formal-kernel certificate.

1.1 Complete polynomial

The unknowns are

$$j, C, z, h, r, v, w, T, c, n, y, p_1, p_2 \in \mathbb{Z}.$$

The six constants $\beta, K_0, M_C, M_F, A_c, B_c$ are the unchanged complete source payloads, with M_F already shifted. Define only polynomial abbreviations:

$$\begin{gathered}
J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \\
X = qW, \quad Y = v, \quad F = (K_0 + W)C - zq_m, \\
R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
k = R + 1 + hE, \quad \eta = c - Yk, \quad L = E(Yk), \\
I = (2r + 1)c^2, \quad D_s = \Delta I^2, \quad F_s = 1 + D_s, \\
Q = \Delta D_s, \quad V = F_s(cT - R) - c, \\
M = 2q^3, \quad U = MH, \quad K_{\mathrm{lo}} = X(2X - 3c) - 2, \\
Z = Un + MK_{\mathrm{lo}} + H(Y - q^3), \\
D_{\mathrm{err}} = (A_cF - q)^2 + (B_cC - q)^2, \\
B_{\mathrm{ud}} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2\big(4(Y\eta)^2 + (kq^3)^2\big), \\
G = 4(B_{\mathrm{ud}} - p_1^2 - p_2^2) - 3, \qquad D_t = 1 + L(L + k), \\
C_1 = (Z + Uk)^2 + 4U(LZ - U), \qquad C_2 = V^2 - (Q - 1)y(2V + y) - 1, \\
g_a = U^2G, \quad d_b = 4U^2D_t, \quad f_c = 16U^2F_s, \quad g_c = g_a + f_c, \\
J_N = (C_1 - g_c)^2 - 4(g_af_c + d_bg_c), \\
N = J_N^2 - 64g_ad_bf_c(C_1 + d_b).
\end{gathered}$$

The single equation is

$$\boxed{P = (N+1)(C_2+1) - 1 = 0.}$$

Sharing D_s avoids recomputing $F_s - 1$ and saves one subtraction. All powers are fixed natural powers; the literal DAG has no divisions or radicals.

1.2 Norm and exact conjunction

For proof only, put $b = 2L + k$, $R_N = Z + Ub$ and $\rho = R_N/U$. The first-carrier identity gives $R_N^2 = C_1 + d_b$. The symmetric norm

$$F_8(S; g, d, f) = \left[(S - g - d - f)^2 - 4(gd + gf + df)\right]^2 - 64gdfS$$

satisfies the polynomial identity

$$F_8(x^2; g, d, f) = \prod_{\epsilon \in \{\pm 1\}^3} \left(x - \epsilon_1\sqrt{g} - \epsilon_2\sqrt{d} - \epsilon_3\sqrt{f}\right).$$

Substitution $S = C_1 + d_b$, $(g, d, f) = (g_a, d_b, f_c)$ gives exactly N. Equivalently,

$$\frac{N}{U^8} = \prod_{\epsilon \in \{\pm 1\}^3} \left(\rho - \epsilon_1\sqrt{G} - 2\epsilon_2\sqrt{D_t} - 4\epsilon_3\sqrt{F_s}\right). \tag{T1}$$

Since N is a square modulo 4, $N \neq -2$. The integer product $(N+1)(C_2+1) = 1$ therefore forces both factors to be 1; it is equivalent to $N = C_2 = 0$ with no sign premise on either residual. This is a tailored use of the standard unit-product device, not a claimed new general conjunction theorem.

1.3 Unconditional signs and square-class separation

The authentic odd $\beta > 1$ makes q even and nonzero. Also H is odd and nonzero, so $U \neq 0$. Since $|W| \geq 4$, X is even with $|X| \geq 8$. Thus $a = (X+1)Y$ is either 0 or has $|a| \geq 7$, giving

$$\Delta = (a+2)^2 - 1 \geq 3, \qquad F_s = 1 + \Delta I^2 \geq 1.$$

Writing $D = XY^2$, we have even D and $D(D+1) \geq 0$, whence $D_t = 1 + D(D+1)k^2$ is a positive odd integer. Also $G \equiv 1 \pmod 4$. If $G < 0$, every factor in (T1) has a nonzero imaginary part from its sole imaginary radical, while $\rho, \sqrt{D_t}, \sqrt{F_s}$ are real. Hence at a zero $G > 0$; $G = 0$ is excluded by its residue.

Write the positive radicands uniquely as

$$G = u_1^2 d_1, \qquad D_t = u_2^2 d_2, \qquad F_s = u_3^2 d_3,$$

where the d_i are positive squarefree and the u_i are positive integers. The numbers u_1, u_2, d_1, d_2 are odd. A vanishing norm factor gives

$$\rho = \epsilon_1 u_1\sqrt{d_1} + 2\epsilon_2 u_2\sqrt{d_2} + 4\epsilon_3 u_3\sqrt{d_3} \in \mathbb{Q}.$$

Distinct squarefree radicals, including $\sqrt{1}$, are linearly independent over $\mathbb{Q}$: sign-change automorphisms in the finite multiquadratic extension, averaged against a chosen character, isolate each coefficient. Group terms with equal squarefree part. A group containing the G term has an odd coefficient sum. A group containing D_t but not G has coefficient sum 2 (mod 4). A remaining group contains only the nonzero F_s term. Thus no irrational square-class group can cancel. An even squarefree class, in particular, could contain only F_s and is also impossible. Consequently all three radicals are integers. With the signs of the vanishing factor, write

$$g = \epsilon_1\sqrt{G}, \qquad \tau = \epsilon_2\sqrt{D_t}, \qquad f = \epsilon_3\sqrt{F_s}; \qquad \rho = g + 2\tau + 4f.$$

Here g is odd; no oddness of F_s was assumed. This is an exact valuation argument within common square classes, not an inference from pairwise distinct radicands or finite tests.

1.4 Complete maps and canonical completeness

From any new zero restore the old coordinates by

$$f_{\mathrm{old}} = f, \qquad p_{3,\mathrm{old}} = (g-1)/2, \qquad n_{\mathrm{old}} = n - g - 4f,$$

retaining every other coordinate. All three restored coordinates are integral. Affineness of Z gives $Z_{\mathrm{old}}/U = 2\tau - b$, making its first carrier zero. The identity $f^2 = F_s = 1 + \Delta I^2$ makes its strong row zero; its auxiliary row is unchanged because the new F_s equals the old f^2. Finally,

$$p_{3,\mathrm{old}}(p_{3,\mathrm{old}} + 1) = (G-1)/4,$$

so its guard is precisely $B_{\mathrm{ud}} = 1 + p_1^2 + p_2^2 + p_{3,\mathrm{old}}(p_{3,\mathrm{old}} + 1)$. This is a literal source zero, after which its ordinary soundness theorem applies. No preliminary CRT, positive orientation or canonical restriction was used.

Conversely, from any old zero delete f, p_3 and put

$$n_{\mathrm{new}} = n_{\mathrm{old}} + (2p_{3,\mathrm{old}} + 1) + 4f_{\mathrm{old}}.$$

The old rows imply $F_s = f_{\mathrm{old}}^2$, $G = (2p_{3,\mathrm{old}} + 1)^2$, and $Z_{\mathrm{old}}/U + b = 2\tau$ with integral $\tau^2 = D_t$. The latter integrality is explicitly proved by the pinned source's monic carrier argument. One factor in (T1) is therefore zero and the auxiliary row is unchanged. This map applies to every old zero, in particular to the exact canonical source34 family supplied from a finite refutation. No extra canonical tuning or simultaneous representation theorem is required.

1.5 Exact coefficient and verification

The degrees of $C_1, F_s, D_t, G, Q, V, C_2, N$ are respectively $22, 12, 18, 18, 18, 17, 36, 120$. Put

$$q_0 = 6\beta j, \quad X_0 = 12q_0 w, \quad Y_0 = v, \quad R_0 = q_0^4 z - 12q_0^3 wC,$$
$$a_0 = X_0 Y_0, \quad U_0 = 8q_0^3 X_0 Y_0.$$

The leading forms are

$$N_{120} = 2^8 U_0^8 R_0^8 q_0^{16} (q_0^8 + 4X_0^2 Y_0^4)^2,$$
$$(C_2)_{36} = 32 a_0^6 r^4 c^8 R_0 y.$$

Their product is the unique degree-156 part, and

$$[j^{106} w^{14} v^{14} z^9 r^4 c^8 y] P = 2^{171} 3^{120} \beta^{106} > 0.$$

The exponent sum is 156, so this is actual degree attainment on the fixed coefficient slice. All thirteen variables and six coefficient ports are live; the unchanged numerals occupy 615883 uncompressed bytes.

The independent normal and optimized audits authenticate the source and numeral bytes, check every named register, multiply all eight signed-root factors, verify the centered norm schedule and lift, prove the coefficient and degree, check literal cost and liveness, and reject all live-gate mutations. Finite checks are not substituted for the square-class proof or the unchanged unbounded source34 compiler/refutation theorem. This endpoint is a new variable/degree/cost tradeoff, not a domination claim over the fourteen-variable degree-88 or fifteen-variable degree-44 constructions.